

Karakteristik Data

(Benih)

Statdes

Statistics

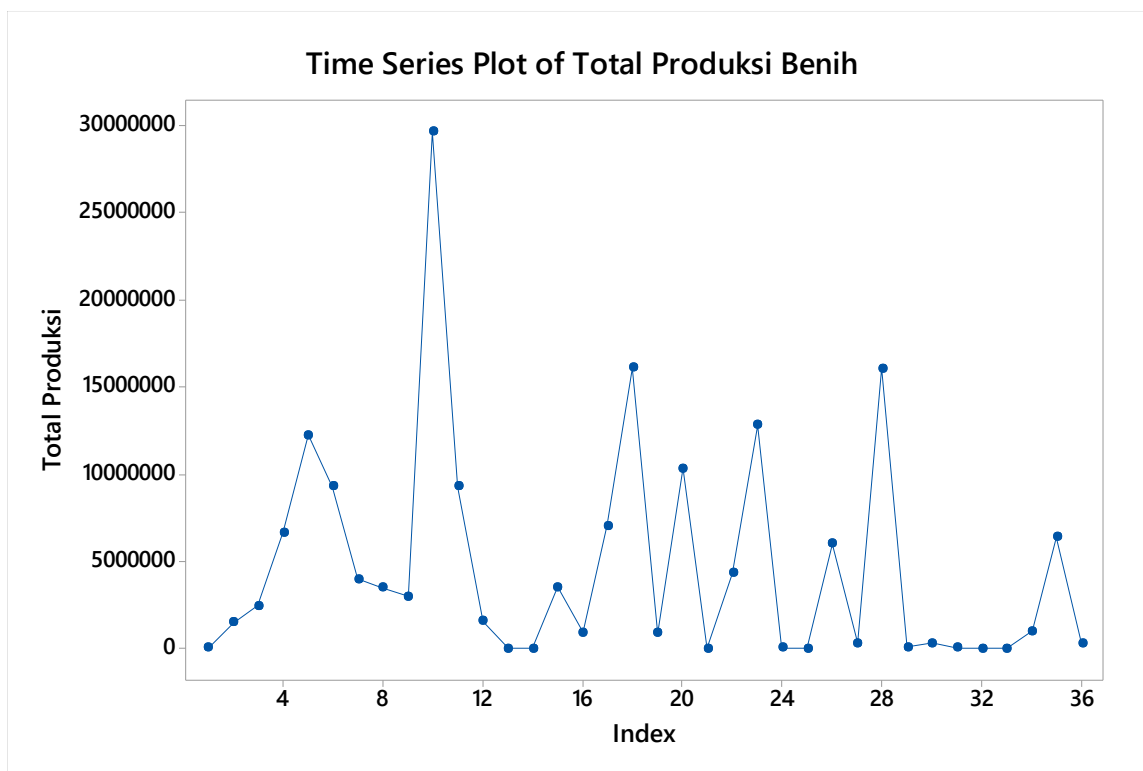
Variable	Mean	StDev	Minimum	Median	Maximum
Total Produksi	4708333	6420059	0	1990000	29640000

```
> summary(ts_data)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	57500	1990000	4708333	6755000	29640000

```
> |
```

Time series plot



(Induk)

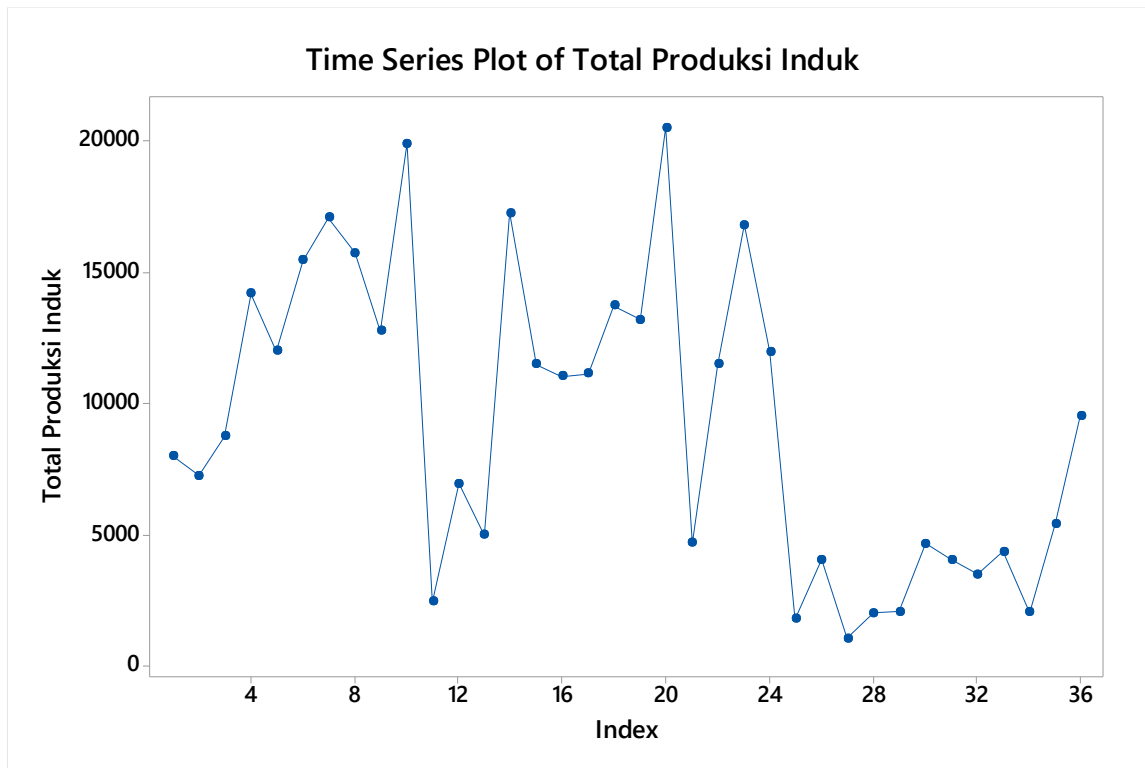
Statdes

Statistics

Variable	Mean	StDev	Minimum	Median	Maximum
Total Produksi Induk	9259	5686	1036	9143	20510

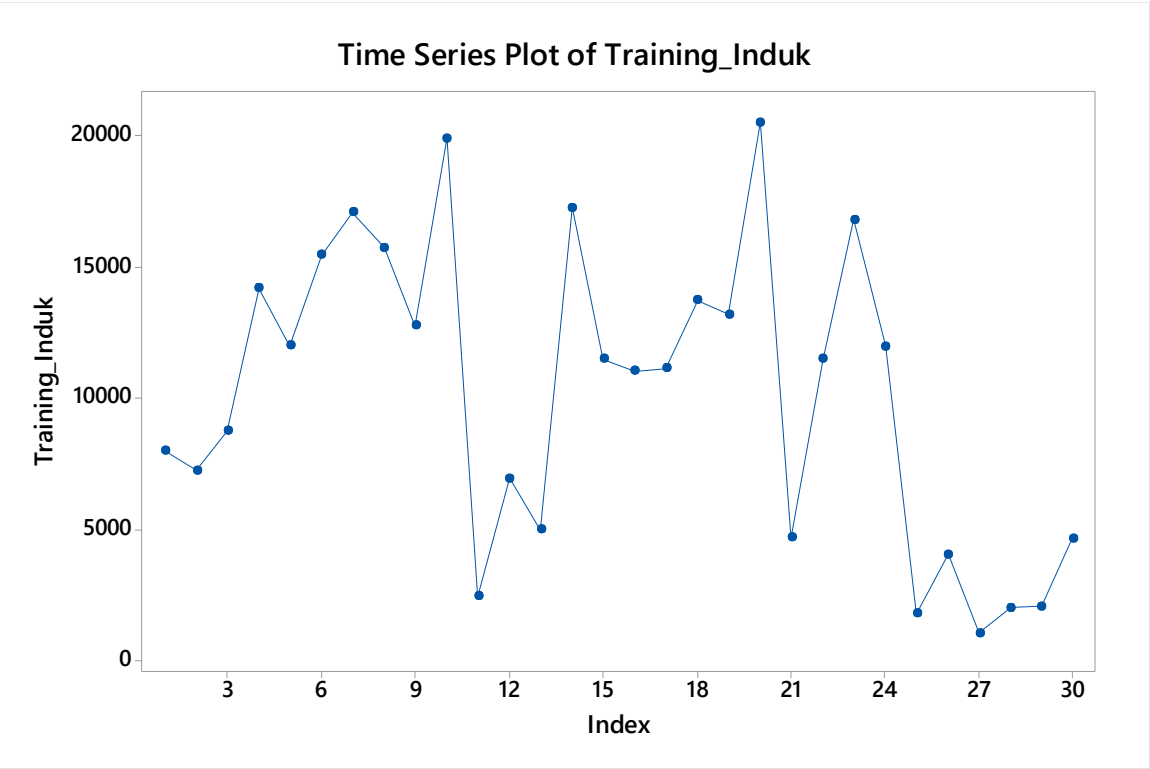
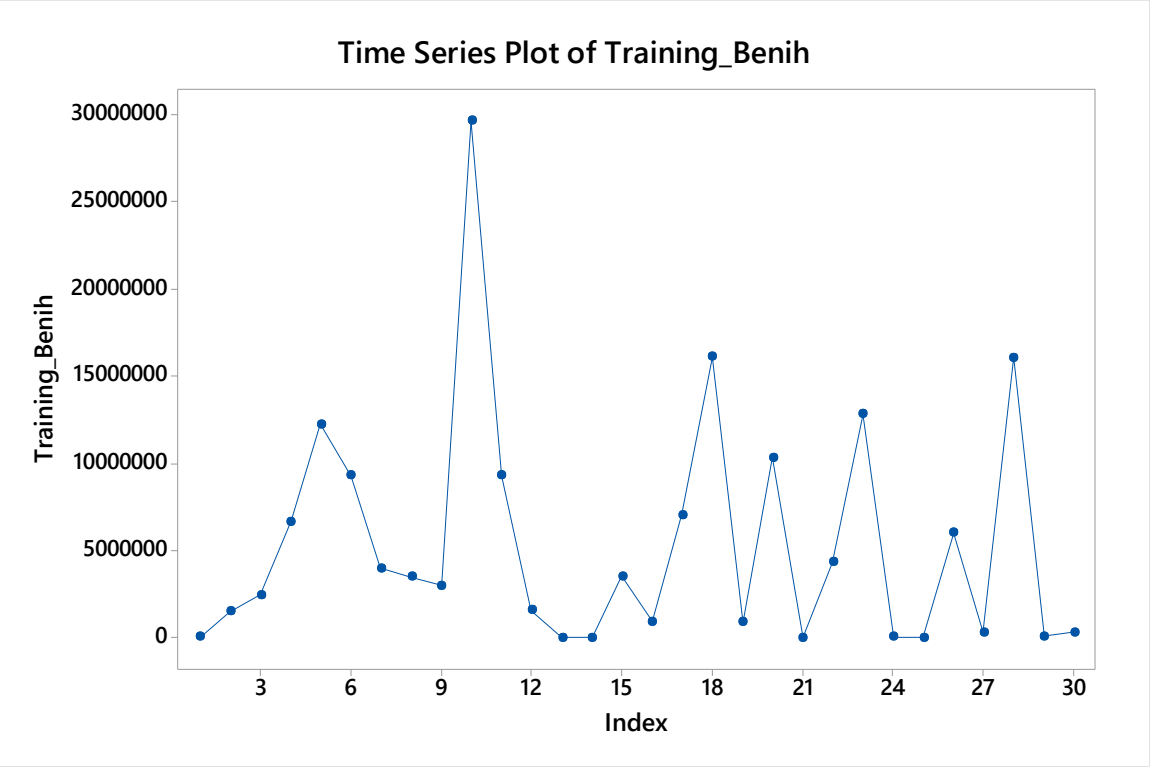
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1036	4280	9142	9259	13328	20510

Time series plot



ARIMA

Time series Plot



Box cox

```
> lambda <- BoxCox.lambda(train)
> lambda      # CATAT ANGKANYA
[1] 1
```

(benih)

```
> lambda      # CATAT ANGKANYA
[1] 1
```

(Induk)

Uji ADF (P Value)

```
Augmented Dickey-Fuller Test

data:  train
Dickey-Fuller = -3.4429, Lag order = 2, p-value =
0.07182
alternative hypothesis: stationary
```

(benih)

```
Augmented Dickey-Fuller Test

data:  train_d1
Dickey-Fuller = -3.6173, Lag order = 2, p-value =
0.04877
alternative hypothesis: stationary
```

(benih d=1)

Augmented Dickey-Fuller Test

```
data: train  
Dickey-Fuller = -3.0363, Lag order = 2, p-value = 0.1776  
alternative hypothesis: stationary
```

```
> train_d1 <- diff(train)  
> autoplot(train_d1) +  
+   ggtitle("Differenced Training Series")  
> adf.test(train_d1)
```

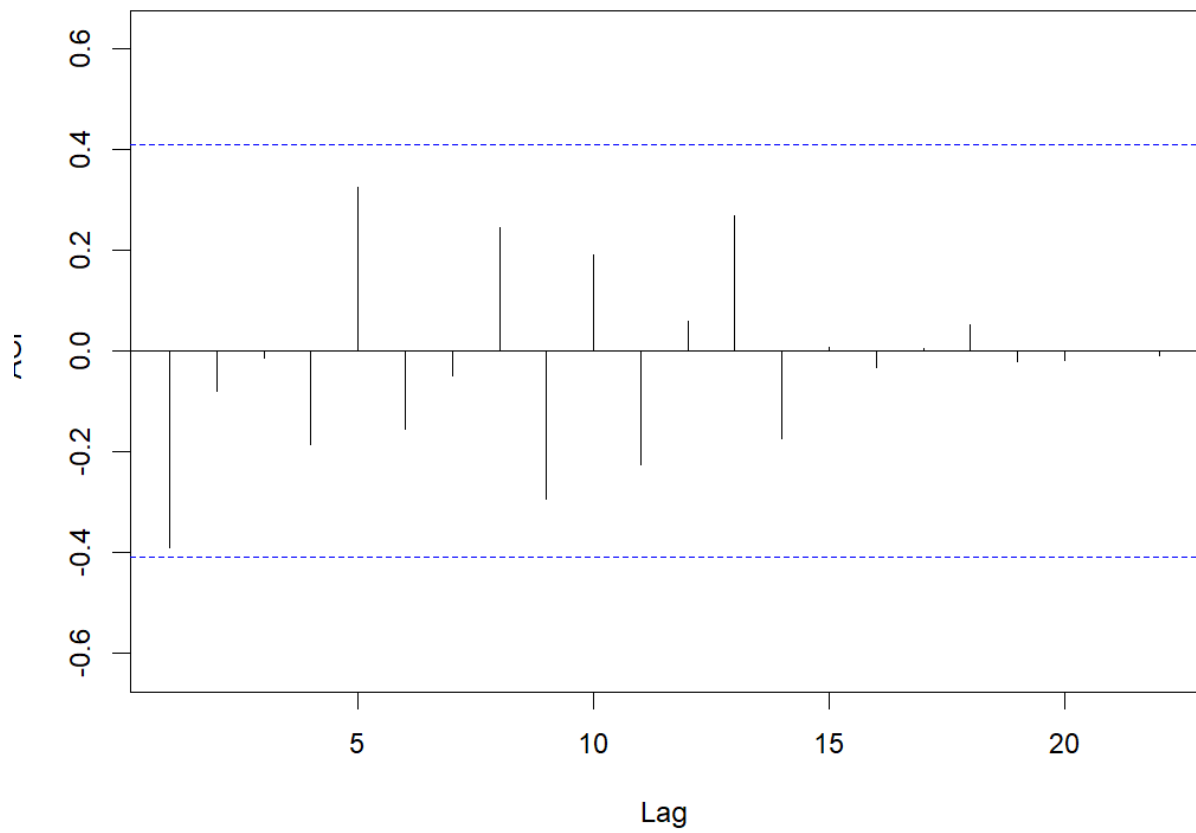
Augmented Dickey-Fuller Test

```
data: train_d1  
Dickey-Fuller = -3.8339, Lag order = 2, p-value = 0.0333  
alternative hypothesis: stationary
```

(Induk)

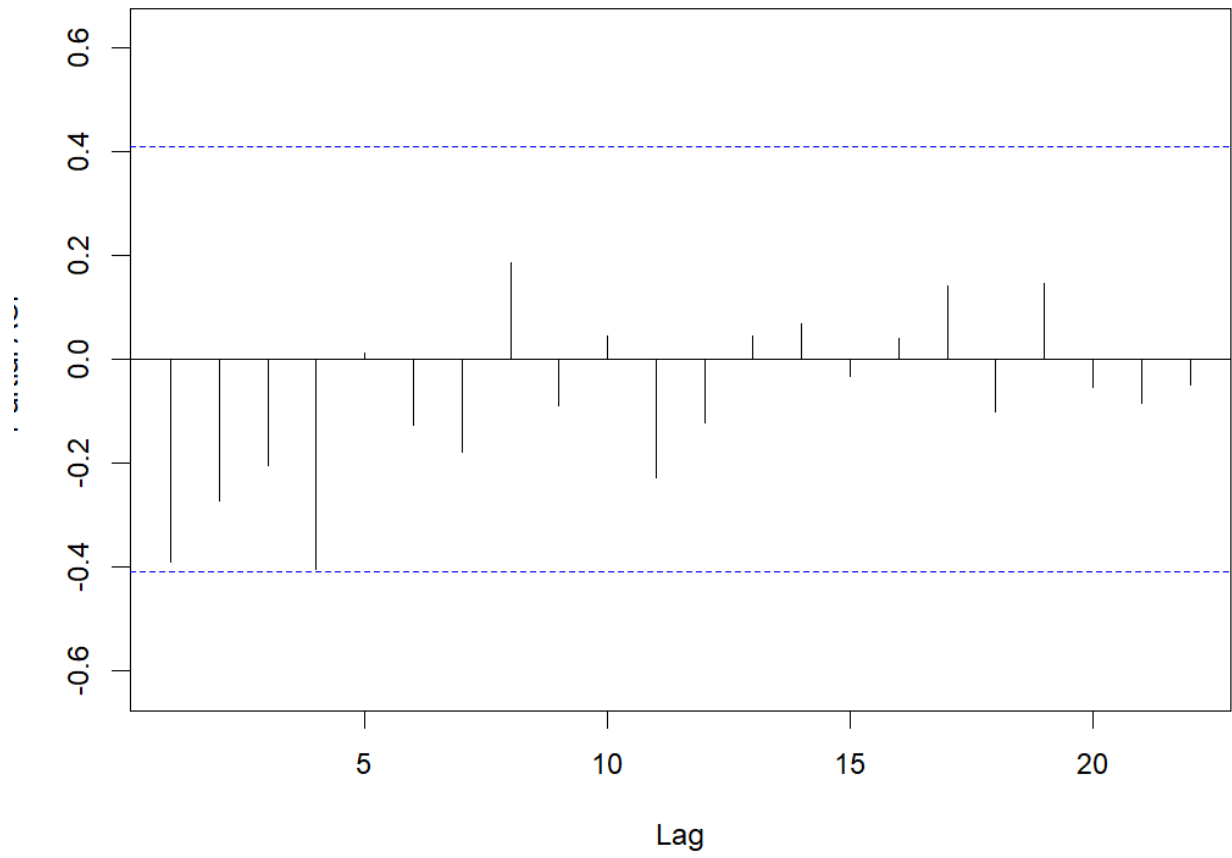
Plot ACF dan PACF

ACF Training



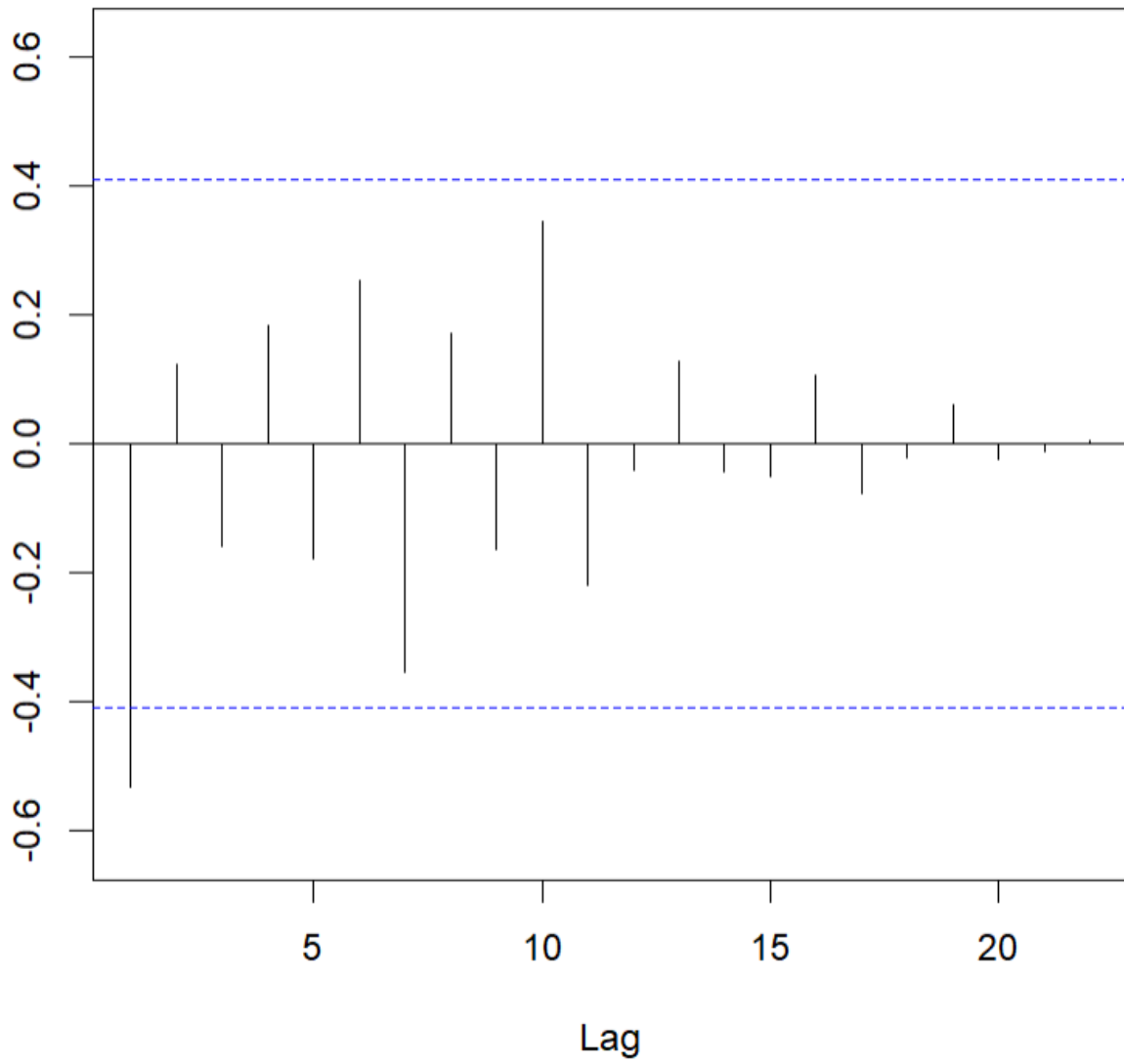
(benih)

PACF Training



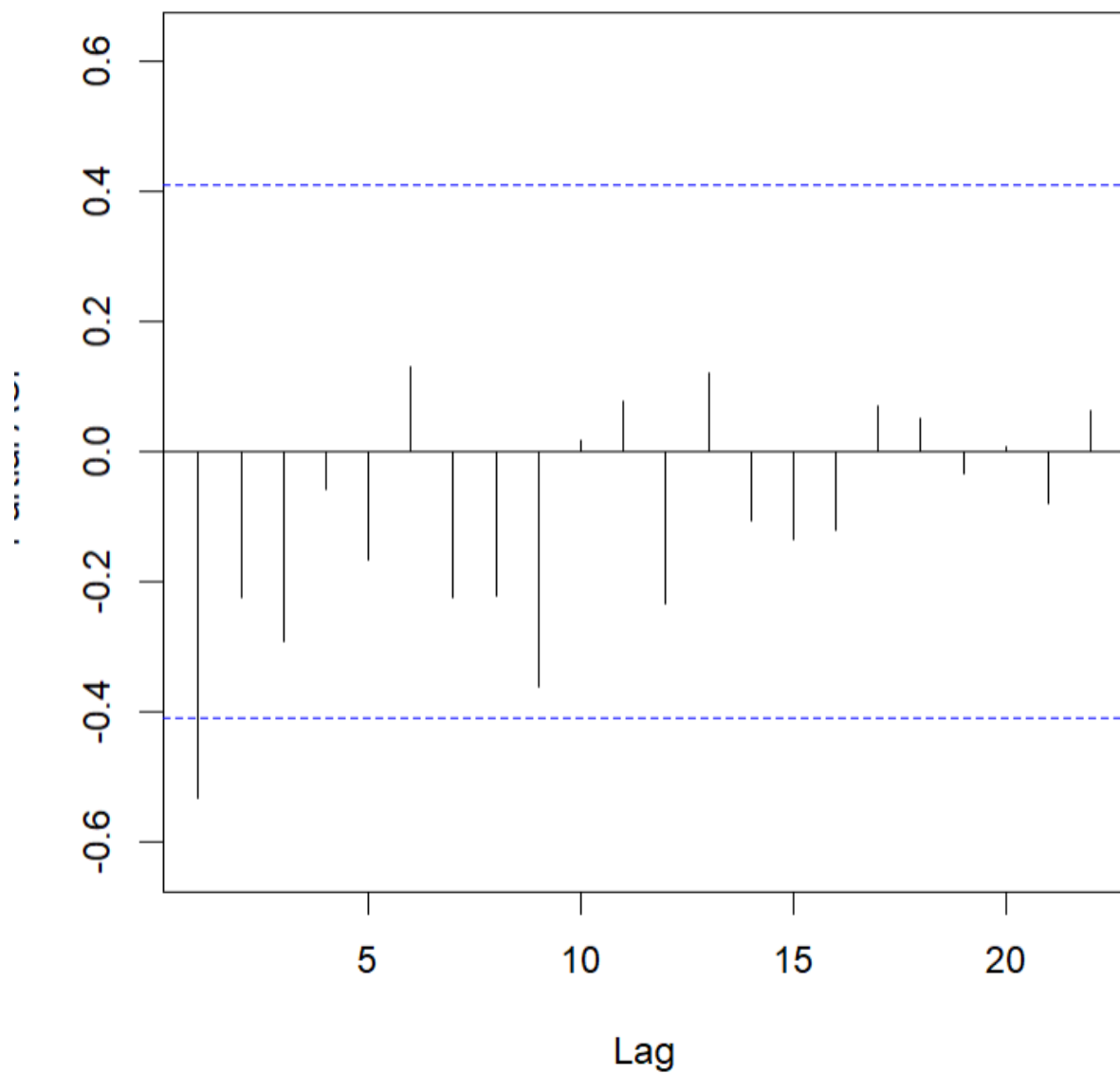
(benih)

ACF Training



(induk)

PACF Training



(induk)

Dugaan Model

```
fit_010 <- Arima(train, order = c(0,1,0))  
fit_110 <- Arima(train, order = c(1,1,0))  
fit_011 <- Arima(train, order = c(0,1,1))  
fit_111 <- Arima(train, order = c(1,1,1))
```

(benih = 0,1,1) dan (induk=0,1,1)

Residual Whitenoise

Box-Ljung test

data: res

X-squared = 8.7922, df = 12, p-value = 0.7206

(benih)

Box-Ljung test

data: res

X-squared = 8.7922, df = 12, p-value =
0.7206

(induk)

Distribusi normal

Exact one-sample Kolmogorov-Smirnov test

data: res_std

D = 0.14189, p-value = 0.6677

alternative hypothesis: two-sided

(benih)

Exact one-sample Kolmogorov-Smirnov test

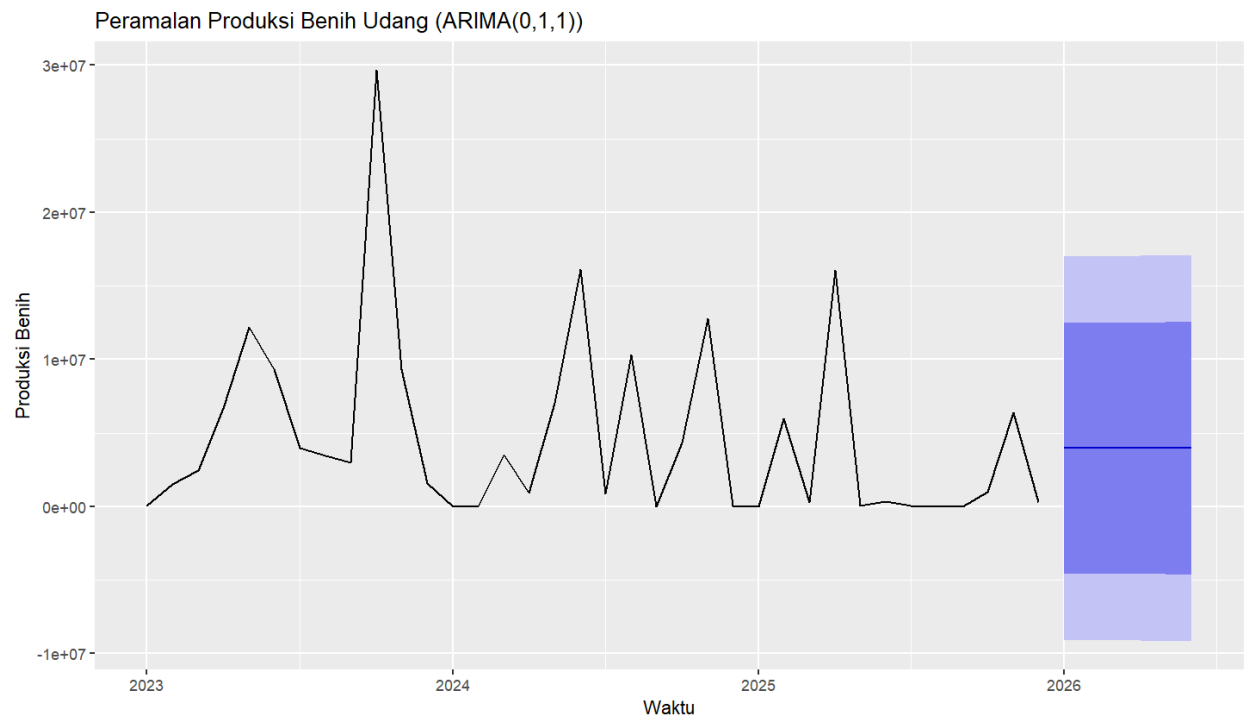
data: res_std

D = 0.14189, p-value = 0.6677

alternative hypothesis: two-sided

(Induk)

Evaluasi model terbaik ARIMA

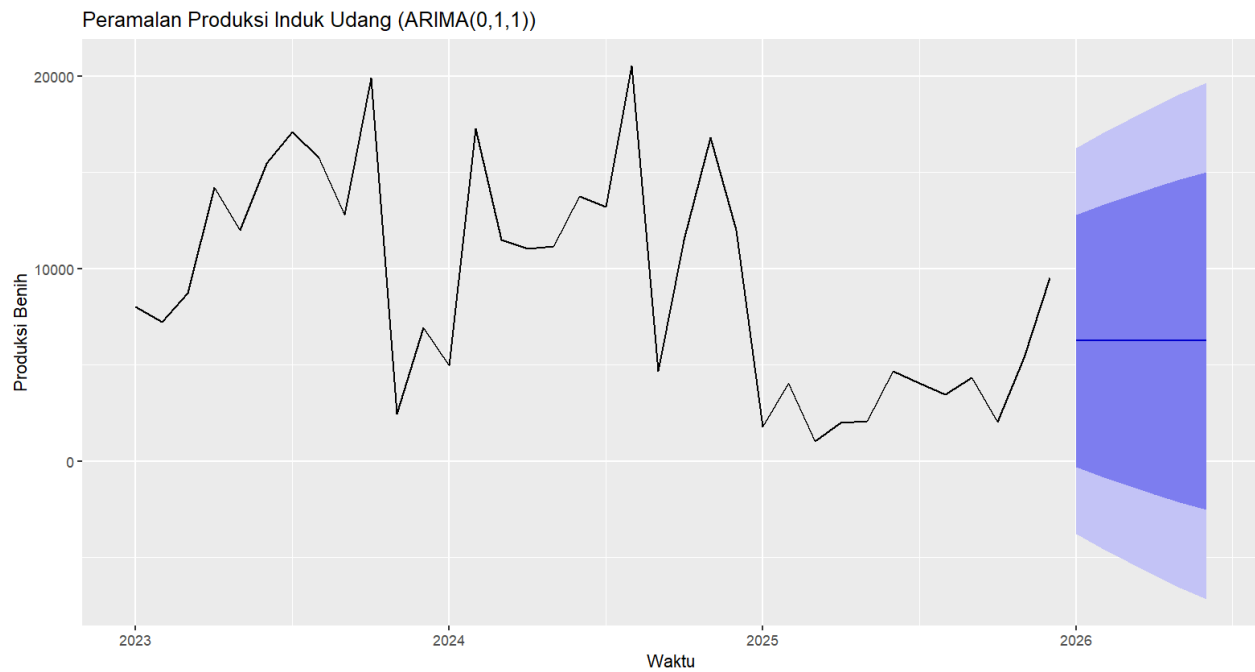


(benih)

```
> forecast_table_final
```

	Periode	Forecast	Lower_95	Upper_95
1	2026.000	3958266	-9057034	16973565
2	2026.083	3958266	-9078242	16994773
3	2026.167	3958266	-9099416	17015947
4	2026.250	3958266	-9120555	17037086
5	2026.333	3958266	-9141660	17058191
6	2026.417	3958266	-9162731	17079262

(benih)



(induk)

```
> forecast_table_final
  Periode Forecast Lower_95 Upper_95
1 2026.000      6243    -3779    16266
2 2026.083      6243    -4544    17031
3 2026.167      6243    -5258    17745
4 2026.250      6243    -5930    18417
5 2026.333      6243    -6568    19054
6 2026.417      6243    -7174    19661
> |
```

(induk)